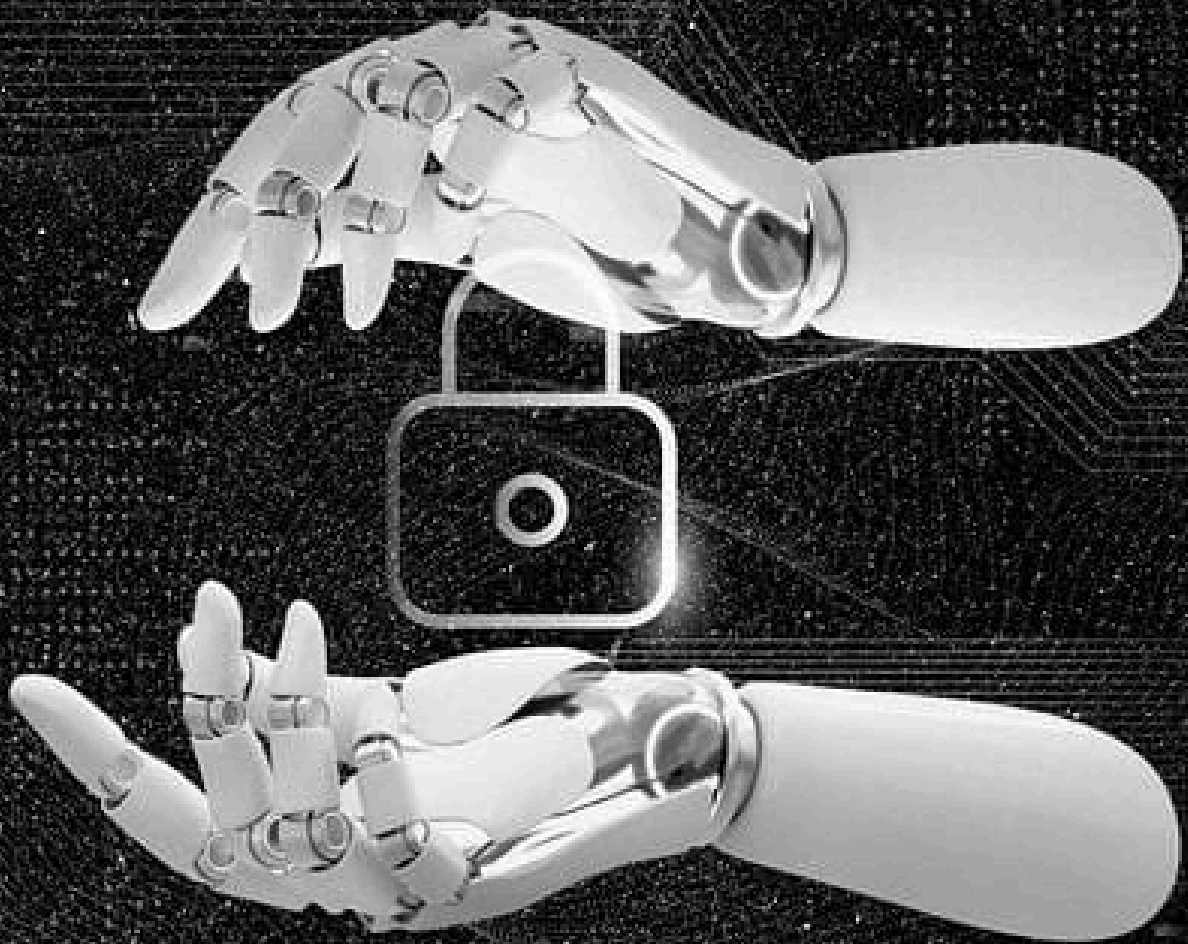
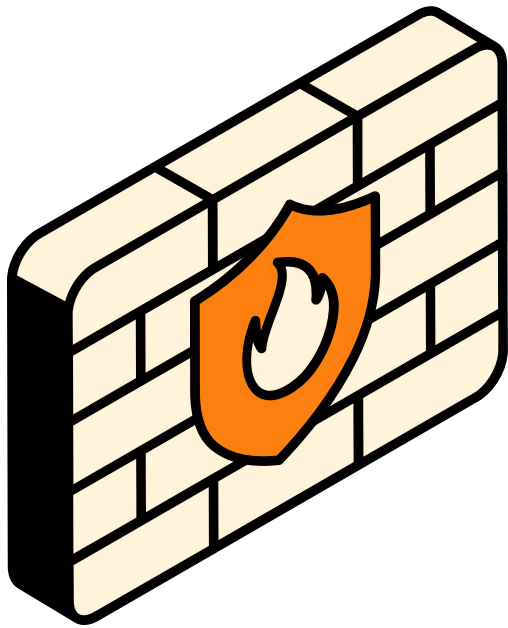




FIREWALL CONFIGURATION AND TRAFFIC FILTERING REPORT

By Sinchana S Kubsad



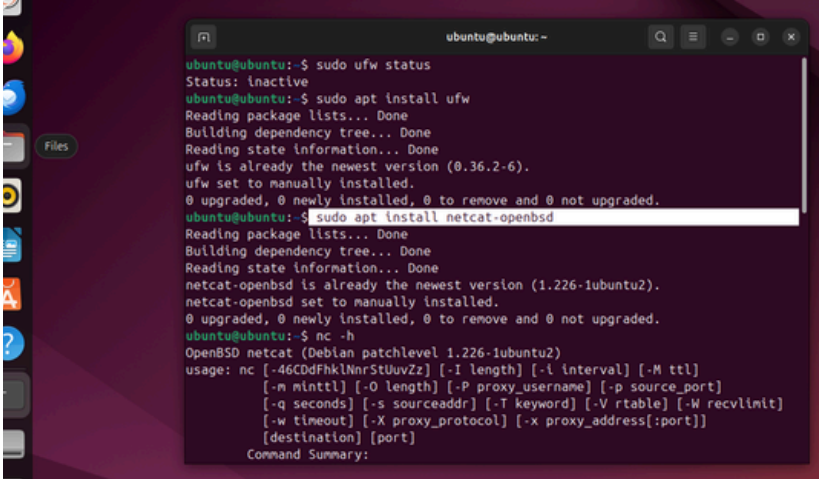
INTRODUCTION

This report documents the configuration and testing performed for Task 4: Firewall Configuration and Traffic Filtering. The primary objective was to demonstrate foundational skills in network security by implementing basic packet filtering rules—specifically, blocking and allowing traffic on specific ports—across two distinct operating environments: Linux (using UFW on Ubuntu) and Windows (using the built-in Windows Defender Firewall). The task successfully confirmed the ability to manage inbound network connections and ensure that unauthorized services (like Telnet/Port 23) are blocked, while necessary services (like SSH/Port 22) are explicitly permitted.

ubuntu LINUX CONFIGURATION AND TESTING (UFW)

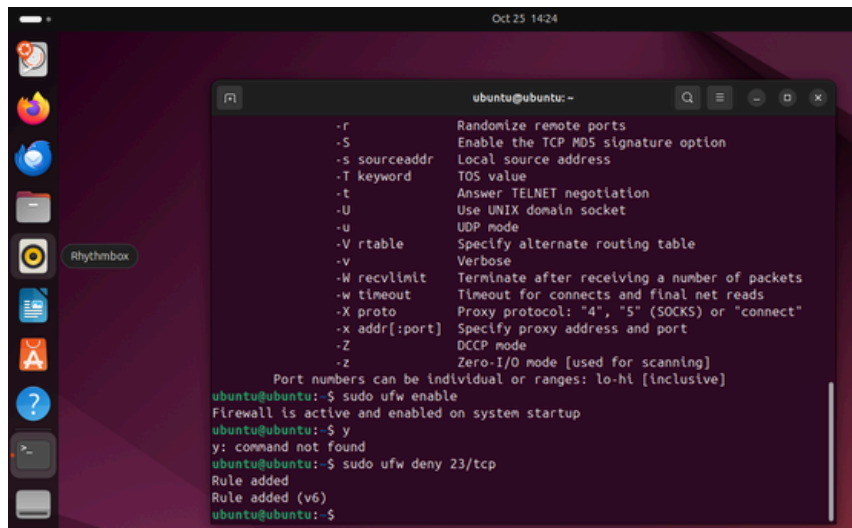
1 - NETCAT INSTALL

This confirms the successful installation of the necessary testing utility, **netcat-openbsd**. The **nc -h** command verifies the program is executable, which is essential for creating the listening service used to validate the firewall rules.



```
ubuntu@ubuntu:~$ sudo ufw status
Status: inactive
ubuntu@ubuntu:~$ sudo apt install ufw
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.2-6).
ufw set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ubuntu:~$ sudo apt install netcat-openbsd
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
netcat-openbsd is already the newest version (1.226-1ubuntu2).
netcat-openbsd set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ubuntu:~$ nc -h
OpenBSD netcat (Debian patchlevel 1.226-1ubuntu2)
usage: nc [-46CDdFhklNnrStUuvZz] [-I length] [-i interval] [-M ttl]
        [-m minttl] [-O length] [-P proxy_username] [-p source_port]
        [-q seconds] [-s sourceaddr] [-T keyword] [-V rtable] [-W recvlimit]
        [-w timeout] [-X proxy_protocol] [-x proxy_address[:port]]
        [destination] [port]
Command Summary:
```

2 - UFW Enable and 23/tcp Deny

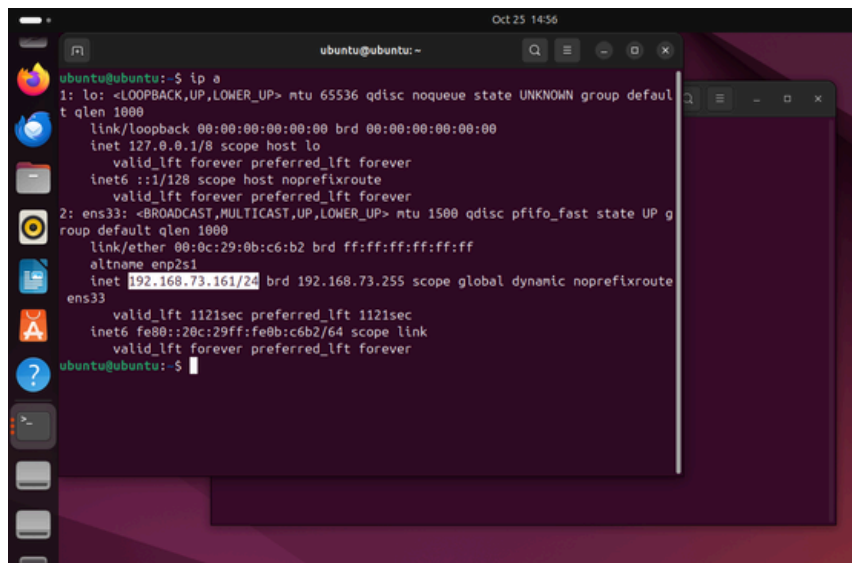
A terminal window on an Ubuntu desktop. The terminal shows the output of the 'man ufw' command, listing various options like -r, -S, -s, -T, -t, -U, -u, -V, -v, -W, -w, -X, -x, -Z. Below this, the user runs 'sudo ufw enable', which returns 'Firewall is active and enabled on system startup'. Then, the user runs 'y: command not found', followed by 'sudo ufw deny 23/tcp', which returns 'Rule added' and 'Rule added (v6)'.

```
ubuntu@ubuntu:~$ man ufw
-r Randomize remote ports
-S Enable the TCP MD5 signature option
-s sourceaddr Local source address
-T keyword TOS value
-t Answer TELNET negotiation
-U Use UNIX domain socket
-u UDP mode
-V rtable Specify alternate routing table
-v Verbose
-W recvlimit Terminate after receiving a number of packets
-w timeout Timeout for connects and final net reads
-X proto Proxy protocol: '4', '5' (SOCKS) or 'connect'
-x addr[:port] Specify proxy address and port
-Z DCCP mode
-z Zero-I/O mode [used for scanning]

Port numbers can be individual or ranges: lo-hi [inclusive]
ubuntu@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
ubuntu@ubuntu:~$ y
y: command not found
ubuntu@ubuntu:~$ sudo ufw deny 23/tcp
Rule added
Rule added (v6)
ubuntu@ubuntu:~$
```

This image shows the initial activation of the **UFW firewall**, confirming its state is active. Immediately following, the **DENY 23/tcp** rule is added. This rule explicitly prevents any inbound connection attempts to the insecure Telnet service, establishing the host-based security perimeter.

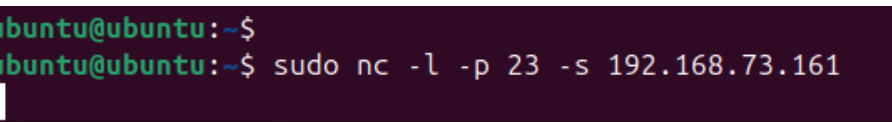
3 - IP addresses

A terminal window on an Ubuntu desktop. The user runs the 'ip a' command, which displays detailed network interface information. The output shows the loopback interface 'lo' with IP 127.0.0.1 and the ethernet interface 'ens33' with IP 192.168.73.161. The IP 192.168.73.161 is highlighted in the original image.

```
ubuntu@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:0b:c6:b2 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.73.161/24 brd 192.168.73.255 scope global dynamic noprefixroute
        valid_lft 1121sec preferred_lft 1121sec
    inet6 fe80::20c:29ff:fe0b:c6b2/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$
```

The **ip a** command was used to identify the host machine's non-loopback network address (**192.168.73.161**). This external-facing IP was necessary for performing a remote connection test, which is required to ensure the firewall rule is being enforced against traffic originating outside the local machine (bypassing the loopback exception).

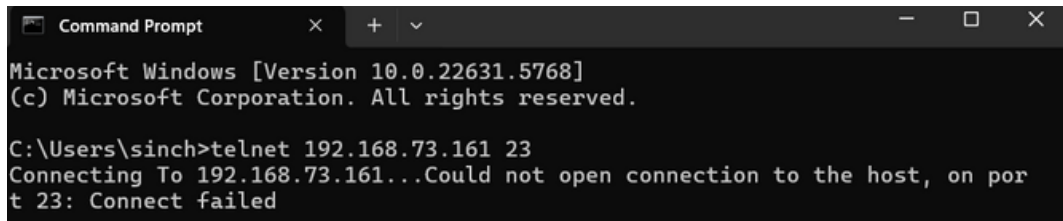
4 - Netcat Listener

A terminal window on an Ubuntu desktop. The user runs the command 'sudo nc -l -p 23 -s 192.168.73.161' to set up a netcat listener on port 23 using the IP 192.168.73.161.

```
ubuntu@ubuntu:~$ sudo nc -l -p 23 -s 192.168.73.161
```

The Netcat utility is started in listening mode (-l) on Port 23, explicitly binding to the host's network IP (-s **192.168.73.161**). The blinking cursor confirms the simulated Telnet service is active and waiting for an inbound connection attempt, setting the stage for the firewall test.

5 - Remote Validation of Firewall Denial (Telnet Block)

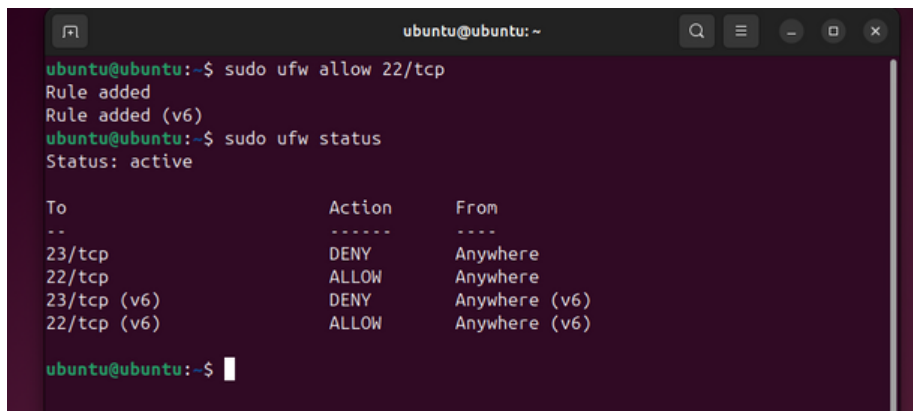


```
Microsoft Windows [Version 10.0.22631.5768]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sinch>telnet 192.168.73.161 23
Connecting To 192.168.73.161...Could not open connection to the host, on port
23: Connect failed
```

This screenshot provides the critical evidence. The connection attempt from the external Windows machine, using Telnet to reach the Linux host on Port 23, resulted in a "**Connect failed**" error. This confirms that the previously implemented **UFW rule (sudo ufw deny 23/tcp)** is correctly enforced against network traffic, successfully blocking the specified port.

6 - Configuring SSH Access and Rule Verification

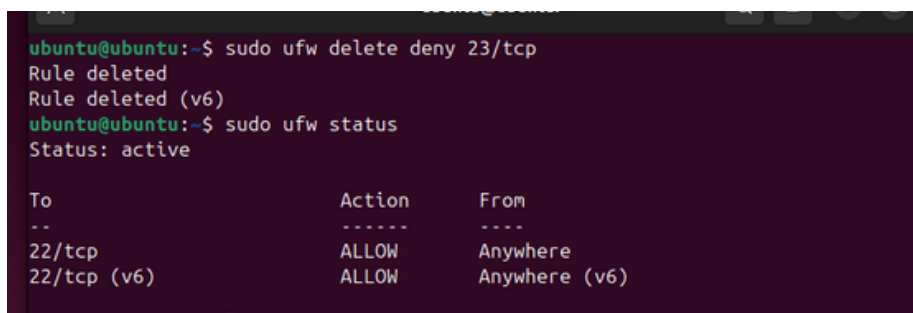


```
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ sudo ufw allow 22/tcp
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
23/tcp DENY Anywhere
22/tcp ALLOW Anywhere
23/tcp (v6) DENY Anywhere (v6)
22/tcp (v6) ALLOW Anywhere (v6)
```

The command **sudo ufw allow 22/tcp** creates the explicit exception needed for secure remote administration **via SSH**. The subsequent **sudo ufw status** command verifies the rule is active and shows the new state of the firewall, listing **ALLOW 22/tcp (SSH)** for both IPv4 and IPv6 traffic. This dual-stack configuration confirms that necessary services are accessible while the overall default policy remains restrictive.

7 - Rule Deletion

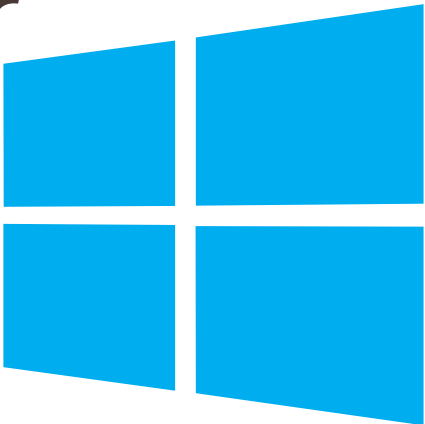


```
ubuntu@ubuntu:~$ sudo ufw delete deny 23/tcp
Rule deleted
Rule deleted (v6)
ubuntu@ubuntu:~$ sudo ufw status
Status: active

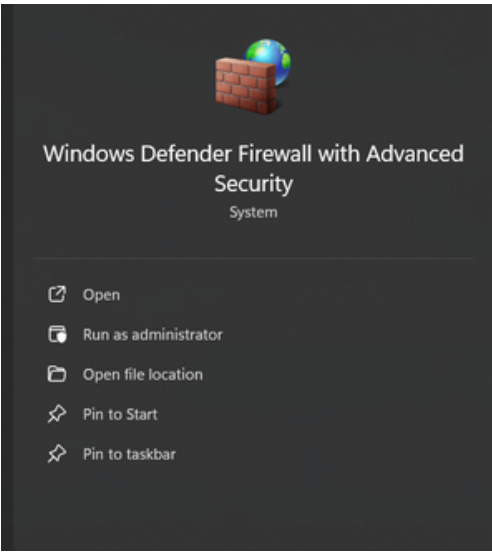
To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
```

To adhere to best practices and restore the firewall to its long-term operational state, the temporary **DENY 23/tcp** rule was safely **removed using the ufw delete** command. This ensures the firewall only maintains the essential ALLOW 22/tcp (SSH) rule, following the Principle of Least Privilege.

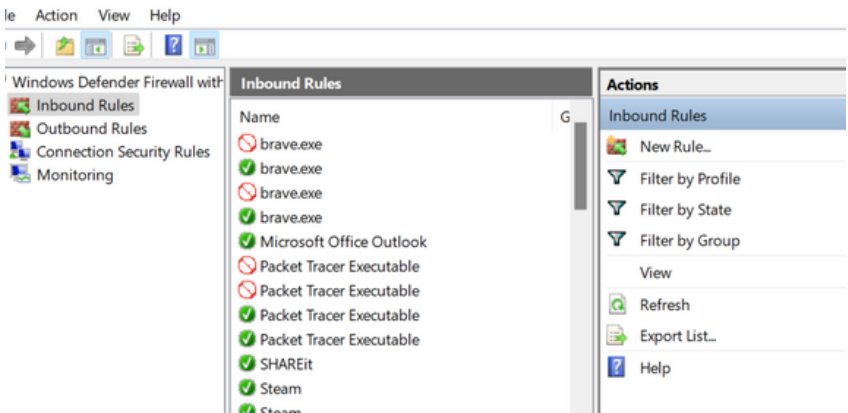
Windows Defender Firewall Configuration and Testing



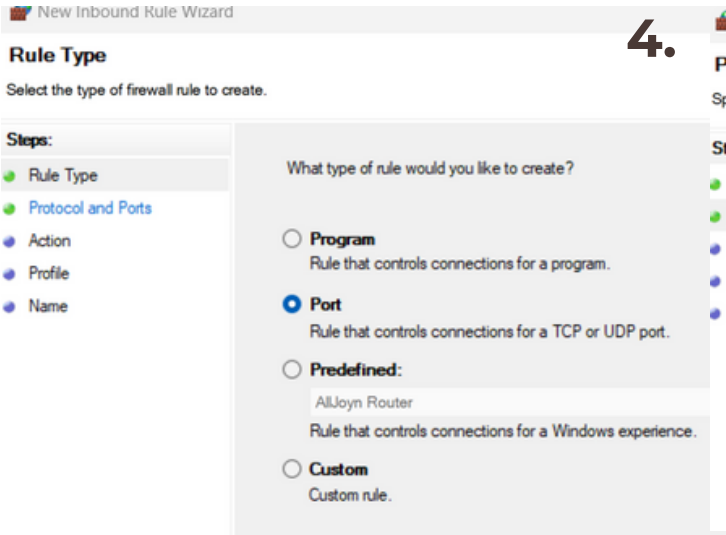
1.



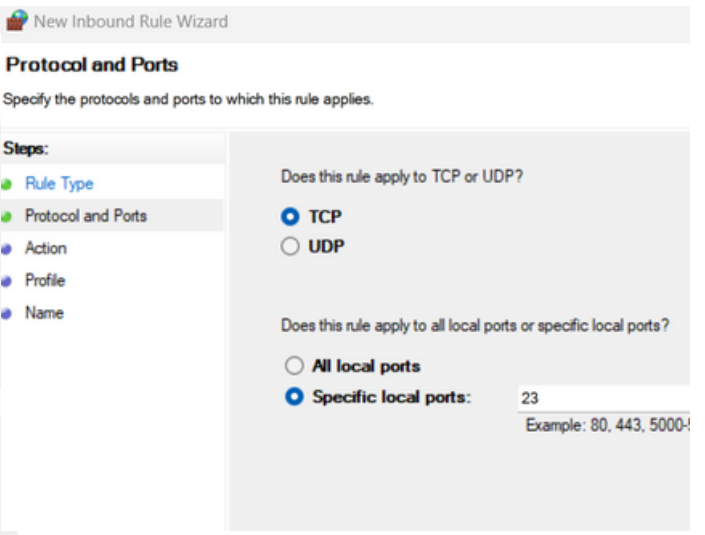
2.



3.



4.



5.

ard

rd

when a connection matches the conditions specified in the rule.

is rule applies.

What action should be taken when a connection matches the specified conditions?

☐ **Allow the connection**
This includes connections that are protected with IPsec as well as those are not.

☐ **Allow the connection if it is secure**
This includes only connections that have been authenticated by using IPsec. Connections will be secured using the settings in IPsec properties and rules in the Connection Security Rule node.

☒ **Block the connection**

When does this rule apply?

☒ **Domain**
Applies when a computer is connected to its corporate domain.

☒ **Private**
Applies when a computer is connected to a private network location, such as a home or work place.

☒ **Public**
Applies when a computer is connected to a public network location.

7.

New Inbound Rule Wizard

Name

Specify the name and description of this rule.

Steps:

- Rule Type
- Protocol and Ports
- Action
- Profile
- Name

Name: block_telnet_23

Description (optional): task 4

8.

Action View Help

Windows Defender Firewall with

- Inbound Rules
- Outbound Rules
- Connection Security Rules
- Monitoring

Inbound Rules	
Name	Group
block_telnet_23	G
brave.exe	
brave.exe	
brave.exe	
brave.exe	
Microsoft Office Outlook	
Packet Tracer Executable	
Packet Tracer Executable	
Packet Tracer Executable	
Packet Tracer Executable	
SHAREit	
Steam	
Steam	
VMware Authd Service	
VMware Authd Service (private)	
@{5A894077.McAfeeSecurity_2.1.68.0_x64_...	...
@{Microsoft.Win32WebViewHost_10.0.220...	...
@{Microsoft.Windows.CloudExperienceHo...	...
@{Microsoft.Windows.Search_1.16.0.22000...	...
@{Microsoft.Windows.StartMenuExperien...	...
@{MicrosoftWindows.LKG.DesktopSpotlig...	...

Actions

Inbound Rules

- New Rule...
- Filter by Profile
- Filter by State
- Filter by Group
- View
- Refresh
- Export List...
- Help

block_telnet_23

- Disable Rule
- Cut
- Copy
- Delete
- Properties
- Help

9.

```
Microsoft Windows [Version 10.0.22631.5768]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sinch>telnet 127.0.0.1 23
Connecting To 127.0.0.1...Could not open connection to the host, on port 23: Connect failed

C:\Users\sinch>|
```

1: Windows Defender Firewall Console Access.

The Windows Defender Firewall with Advanced Security management console is the administrative interface used to configure host-based rules. Accessing this tool is the mandatory first step for implementing custom packet filtering policies on the Windows operating system.

2: Selection of Inbound Rules.

The Inbound Rules node is selected. This area manages policies that filter traffic attempting to enter the host machine, which is the necessary location for creating the DENY rule for the Telnet port.

3: Selecting Rule Type (Port).

The New Inbound Rule Wizard is initiated. The Port rule type is chosen, as the objective is to filter traffic based on the TCP destination port number (23), rather than by a specific program or IP address.

4: Defining Protocol and Port for Filtering.

The rule is defined to apply to the TCP protocol, which is used by Telnet. The traffic is explicitly filtered on Specific local ports, with 23 entered as the target. This ensures the rule is narrow in scope, affecting only the designated insecure service.

5: Defining the Rule Action.

The desired action for any traffic matching the TCP Port 23 criteria is set to Block the connection. This configuration ensures that any inbound packets attempting to access the Telnet service are immediately dropped by the firewall, fulfilling the denial objective.

6: Profile Selection for Universal Enforcement.

All three network profiles (Domain, Private, and Public) are selected. This action ensures that the denial rule remains enforced regardless of the network the host machine is currently connected to (e.g., corporate network, home network, or a public hotspot).

7: Finalizing the Rule Name.

The rule is assigned a descriptive name (e.g., BLOCK_TELNET_23) and an optional description. Clear naming is a critical step in professional firewall management, ensuring that the rule's purpose can be immediately identified for future auditing or maintenance.

8: Local Validation of Windows Firewall Denial. 🔑

This screenshot confirms the rule's successful enforcement. By attempting to connect to the local machine's Port 23 (127.0.0.1) after the rule was created, the immediate "Connect failed" message proves the Block the connection policy is active and successfully preventing traffic from accessing the specified port.

Conclusion and Traffic Filtering Summary

Summary: How Firewall Filters Traffic

A firewall operates as a digital security gatekeeper, applying a set of predefined rules to inspect all incoming and outgoing data packets. Its core function is to implement the Principle of Least Privilege at the network layer. It primarily filters traffic by examining:

Source/Destination IP Addresses: Where the data is coming from and where it's going.

Port Numbers: Which specific application the data is intended for (e.g., Port 22 for SSH, Port 80 for HTTP).

Protocol: Whether the traffic is using TCP or UDP.

If a packet matches an explicit Allow rule (like the one implemented for Port 22), it is permitted to pass. If it matches a Deny rule (like the block on Port 23), or if it matches no rule at all (due to the restrictive default policy), the packet is immediately dropped (discarded). This ensures only necessary and approved communication can reach the host machine.

Interview Q&A

1. What is a Firewall?

A firewall is essentially a digital security guard that monitors and controls all the traffic flowing into and out of a computer or network. It works by checking every data packet against a specific set of rules you define. Its job is simple: decide whether to allow the traffic to pass or deny (block/drop) it, ensuring only safe and authorized communication gets through.

2. Difference Between Stateful and Stateless Firewalls

The key difference is memory, or "state."

A Stateless Firewall is simple-minded; it inspects every single data packet individually, in isolation, without remembering if it belongs to an ongoing connection. It checks the packet's source and destination, and that's it.

A Stateful Firewall is much smarter. It keeps track of the "state" of all active connections in a table. For example, if you send an outgoing request to a website, the firewall remembers that request and automatically allows the incoming response (the return traffic) without needing an explicit rule for it. This makes it far more efficient and secure.

3. What are Inbound and Outbound Rules?

These terms just define the direction of the traffic the rule applies to:

Inbound Rules apply to traffic that is trying to enter your computer (like someone trying to connect to your web server).

Outbound Rules apply to traffic that is trying to leave your computer (like an application trying to send data out to the internet).

4. How Does UFW Simplify Firewall Management?

UFW (Uncomplicated Firewall) makes managing the Linux firewall much easier by removing the complexity of the underlying system, iptables.

It lets you use simple, human-readable commands like `sudo ufw allow 22` instead of long, confusing technical strings.

It automatically handles things like setting up both IPv4 and IPv6 rules for you, so you don't have to duplicate every step.

It simplifies the process of setting a secure "deny by default" policy, allowing you to focus only on the few things you need to explicitly allow.

5. Why Block Port 23 (Telnet)?

Port 23 is blocked because it is used by the Telnet protocol, which is completely insecure. Telnet sends all data, including your username and password, across the network in plain, unencrypted text.

Any attacker monitoring the network can easily read your login details.

It has been replaced entirely by SSH (Secure Shell) on Port 22, which encrypts all communication and is the standard for secure remote access.

6. What are Common Firewall Mistakes?

Leaving the Door Open: The most common mistake is not setting a "deny all incoming" default policy, which means any port you haven't explicitly blocked is actually open.

Too Broad Rules: Creating rules that allow traffic from any source IP address (using the "ANY" option) when the access should be limited to specific IPs.

Forgetting to Delete Test Rules: Failing to remove temporary test rules or old rules, which can later cause unexpected connection issues or unnecessarily expose a port.

No Documentation: Creating rules without clear names or comments, making them impossible to audit or clean up later.

7. How Does a Firewall Improve Network Security?

A firewall is essential because it is the first line of defense that enforces the Principle of Least Privilege for network access.

It reduces the attack surface by closing down tens of thousands of unused ports, minimizing the ways an attacker can probe your system.

It controls unauthorized access, ensuring that only authenticated protocols (like SSH) and verified users can interact with your host.

It prevents unwanted programs or malware from communicating with the outside world (using outbound rules).

8. What is NAT in Firewalls?

NAT (Network Address Translation) is a feature commonly found in network firewalls (like your home router).

Its primary job is to let many devices inside a private network (like your phone, laptop, and tablet, all with private IP addresses) share a single public IP address when communicating with the internet.

As a security benefit, NAT hides the internal IP addresses and the structure of your private network from the outside world, adding a layer of defense by making it harder for external systems to directly target your internal devices.